2026 Notebook

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The purpose of this yearly notebook is to document large scale progress in projects, any significant mistakes, important derivations, nice tricks that I think will serve me into the future, etc.

September

New Methods

M-estimation, Z-estimation, sandwich variance

Influence Functions, Efficient Influnece Functions, connections between IF/EIF and estimating functions

Mistakes I made

October

Debiased Machine Learning (BIP Project)

Debiased / Double Machine Learning (DML) is a semiparametric estimation method that came out of the economics literature that allows us to obtain an unbiased and efficient estimate for a low-dimensional target parameter ψ (perhaps to do statistical inference) while accounting for high-dimensional nuisance parameters η. However, they have become useful in biostatistics. Example: Jinbo paper.

Q1. What is the simplest framework that we are familiar with: Partial Linear Regression (generalized additive model type thing?) and a connection to the Frisch-Waugh-Lovell theorem we know from linear models.

$$Y \sim X + f(V)$$

This function is infinite-dimensional (a thought exercise and an informal proof in linear algebra).

Q2. How do we estimate this infinite-dimensional nuisance with the data? Machine Learning!

The double part of this method is that we use two separate machine learning models for the partial Y and partial X (notated as D in these papers)

Thought: What might go wrong with just fitting a machine learning model on this equation?

Q3. Machine learning methods are a black box… how do we confirm we obtain an unbiased estimate of our target parameter(s)?: Neyman Orthogonality. There is some detail that I don't know if I can go into in time…

Some references:

1) The canonical paper: Double/Debiased Machine Learning for Treatment and Structural Parameters1 Victor Chernozhukov†, Denis Chetverikov‡, Mert Demirer† Esther Duflo†, Christian Hansen§, Whitney Newey†, James Robins
2) A simpler introduction: https://arxiv.org/abs/2504.08324

I will motivate it using RCT data: https://bingkaiwang.github.io/RCT_Bench-main/website/. Unless I am using a noncollapsible model, adjusting for all covariates as my nuisance and using penalized machine learning should still help in semiparametric efficiency gains compared to non-adjustment or simple adjustment (not entirely convinced by that last point)